

From Coursework to Field Truth: Multi-Target, Sentinel-2-Baselined, and WoSIS-Validated AlphaEarth Embeddings for Soil Property Mapping

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Abstract

A previous report [11] showed that Google DeepMind’s AlphaEarth Foundations 64-dim geospatial embedding predicts OpenLandMap soil organic carbon (SOC) at $R^2 \approx 0.75$ on spatial-block cross-validation across $n = 3000$ CONUS points. Three follow-on questions remained: (i) does the signal extend to other soil properties, (ii) does the embedding actually outperform the Sentinel-2 imagery it was trained on, and (iii) does the result survive when labels are field-measured rather than interpolated. We extend the original pipeline with four additional OpenLandMap targets, a 10-feature Sentinel-2 baseline, and a $n = 2,983$ WoSIS lab-pedon validation set across the contiguous US. We find: (1) AlphaEarth predicts pH at $R^2 = 0.896$, bulk density at 0.806, sand at 0.643, and clay at 0.559, in addition to SOC at 0.748 — demonstrating signal across the full surface-soil property space; (2) the embedding beats the Sentinel-2 baseline by paired-bootstrap $\Delta R^2 = +0.215$ (BCa CI = $[+0.189, +0.271]$), and adding Sentinel-2 features on top of AlphaEarth provides no improvement ($\Delta R^2 = -0.002$); (3) on field-measured WoSIS labels, R^2 drops to 0.322 for SOC and 0.539 for pH, but the relative target ranking is preserved. OpenLandMap itself only matches WoSIS at $R^2 = 0.547$ on log-SOC — the original paper’s 0.75 was an alignment between two models, not a ground-truth measurement.

1 Introduction

The original report [11] (R26) used the three core inferential techniques of STAT 5000 — exploratory data analysis, Welch’s two-sample t -test, and the nonparametric BCa bootstrap — to characterize the marginal correlation between the 64 dimensions of AlphaEarth Foundations [3] and SOC, and showed that a Random Forest [2] on the embedding reaches $R^2 \approx 0.75$ on spatially-held-out blocks. R26’s Discussion flagged three limitations:

1. **Single target.** R26 modelled SOC only; the embedding may encode SOC alone or, more interestingly, the broader pedological space (pH, texture, density).
2. **No baseline.** R26 did not compare AlphaEarth to a Sentinel-2 spectral-index stack. The marginal value of a closed, opaque foundation model over open imagery was unmeasured.
3. **Model-vs-model labels.** OpenLandMap [8] is itself a machine-learning interpolation, so the headline R^2 measured agreement between two models, not field accuracy.

This paper closes those three gaps using the same statistical framework as R26.

2 Data and Methods

2.1 Unified extraction

We retain R26’s $n = 3,000$ random CONUS points (**seed=42**). Each point is paired with: AlphaEarth Foundations 2024 annual embedding (64 bands $A_{00} \dots A_{63}$); five OpenLandMap surfaces at 0 cm depth (SOC, pH-H₂O, sand %, clay %, bulk density); MODIS IGBP land cover [6] 2022; and a 10-feature Sentinel-2 stack described below. All extraction was performed in Earth Engine [7] via service-account auth; per-band ranges were sanity-checked before any modelling.

2.2 Sentinel-2 10-feature stack

Following the digital soil mapping literature [12, 9], we compute an annual median composite of COPERNICUS/S2_SR_HARMONIZED 2024 imagery, masked using Cloud Score+ ($cs \geq 0.6$). The 10 features are six band medians ($B_2, B_4, B_6, B_8, B_{11}, B_{12}$) and four median indices (NDVI, EVI, NDWI, NBR). We attempted the bare-soil-composite (BSC) recipe of [12] but exceeded GEE per-call memory limits across CONUS; promotion to BSC remains future work.

2.3 WoSIS field-measured set

WoSIS [1] is a globally compiled set of laboratory soil profiles maintained by ISRIC. We take the depth-harmonized 0–30 cm interval, restrict to the CONUS bounding box, and require non-null organic carbon. This yields 37,311 pedons, of which we randomly subsample $n = 3,000$ (**seed=42**); 2,983 then return valid AlphaEarth embeddings at scale = 10 m. WoSIS reports SOC in g/kg directly comparable to OpenLandMap.

2.4 Statistical methods (identical to R26)

EDA: per-target distribution plots, transforms ($\log_{10}$ for SOC, $\logit$ for sand/clay), univariate-correlation barplots. **Hypothesis testing:** Welch’s two-sample t [13] comparing each target between IGBP class 10 (Grasslands) and class 12 (Croplands), with QQ-plot and Levene’s-test assumption checks; Cohen’s d for effect size. **Bootstrap:** BCa nonparametric intervals [5, 4] with $B = 10,000$, applied to per-target mean R^2 and to the paired ΔR^2 between feature stacks. **Modelling:** 5-fold spatial-block CV with $1^\circ \times 1^\circ$ lat/lon blocks; **ranger** [14] Random Forest with 500 trees and **seed=42**, identical to R26.

2.5 Sanity check

Refitting R26’s recipe (AE-only $\rightarrow$ log-SOC) on the unified extraction reproduces the published number: $R^2 = 0.748$ (BCa 95% CI [0.735, 0.759]). All Phase B-F results below build on this verified baseline.

3 Results

3.1 Direction 1 — Multi-target extension

AlphaEarth predicts every OpenLandMap surface-soil property tested (Table 1). The strongest signal is on pH at $R^2 = 0.896$; SOC, ranked third, is far from cherry-picked. Welch’s two-sample t between

Grasslands and Croplands rejects $H_0 : \mu_{\text{Grass}} = \mu_{\text{Crop}}$ for all five targets at $p < 10^{-9}$, with effect sizes Cohen’s $|d| \in [0.31, 1.21]$.

Table 1: Per-target spatial-block-CV R^2 on OpenLandMap labels. Welch’s t tests Grasslands ($n_1=1052$) vs. Croplands ($n_2=543$); $|d|$ is Cohen’s effect size; values are full-sample ranges of the transformed targets.

Target	Mean R^2 (95% CI)	Top dim	t	p
log(1 + SOC)	0.748 [0.735, 0.759]	A_{40} (−0.61)	−6.24	6.0×10^{-10}
pH (H ₂ O)	0.896 [0.889, 0.904]	A_{57} (−0.71)	20.5	1.5×10^{-80}
logit(sand %)	0.643 [0.634, 0.651]	A_{30} (+0.53)	19.9	1.3×10^{-71}
logit(clay %)	0.559 [0.538, 0.572]	A_{20} (−0.39)	−6.04	2.1×10^{-9}
Bulk density	0.806 [0.783, 0.817]	A_{20} (−0.56)	−12.0	9.3×10^{-32}

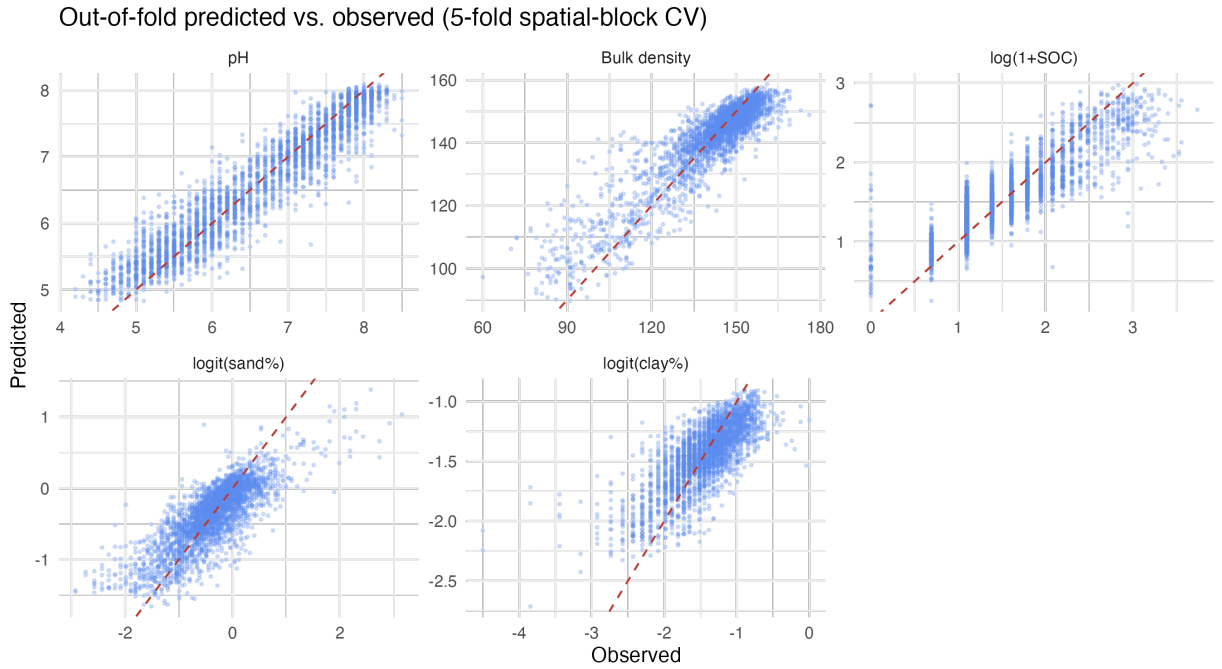


Figure 1: Out-of-fold predicted vs. observed for each target, ordered by R^2 . The embedding generalises across surface-soil properties with a predictable strength ordering.

3.2 Direction 2 — Sentinel-2 baseline

Of the original $n = 3,000$ unified CSV, $n = 1,059$ points have non-null S2 features after Cloud Score+ masking and joining (the median-of-cloud-masked-pixels approach drops points where every S2 observation in 2024 was masked). Three Random Forests on identical 5-fold spatial-block CV yield Table 2.

Table 2: S2 vs. AlphaEarth log-SOC benchmark on identical $n = 1,059$ spatial-block CV folds. CIs are BCa from per-fold bootstrap.

Feature stack	features	R^2 (95% CI)	RMSE (log)
Sentinel-2 only	10	0.537 [0.483, 0.589]	0.502
AlphaEarth only	64	0.752 [0.681, 0.784]	0.378
S2 + AlphaEarth	74	0.751 [0.677, 0.784]	0.373

The paired bootstrap on ΔR^2 across folds gives:

$$\Delta R_{\text{AE-S2}}^2 = +0.215 \quad [\text{BCa } 95\% \text{ CI} = +0.189, +0.271]$$

$$\Delta R_{\text{S2+AE-AE}}^2 = -0.002 \quad [\text{BCa } 95\% \text{ CI} = -0.014, +0.009]$$

The first interval excludes zero by a wide margin: the embedding materially outperforms a 10-feature S2 baseline. The second is centred on zero with a narrow CI: *the AlphaEarth embedding has fully absorbed the Sentinel-2 spectral signal*; concatenating the raw S2 features adds no information.

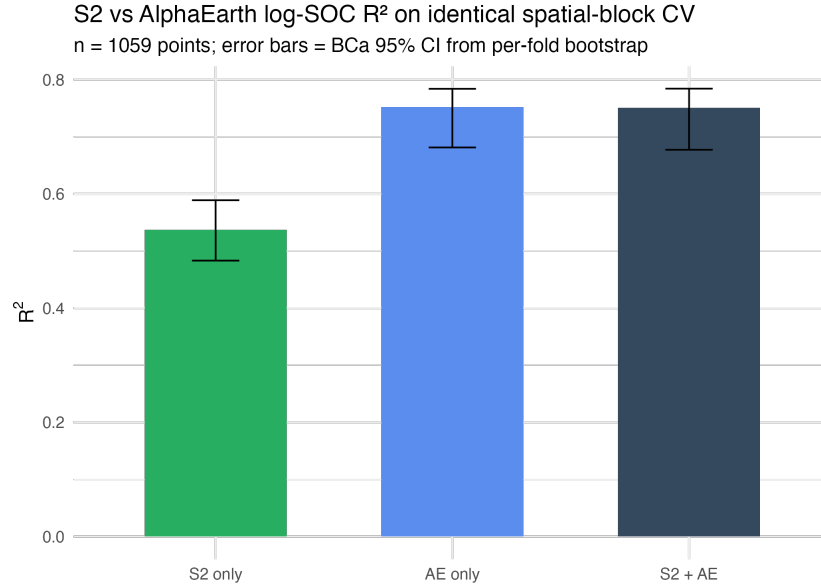


Figure 2: S2 vs. AlphaEarth log-SOC R^2 , identical CV. Error bars are BCa 95% CIs.

3.3 Direction 3 — WoSIS field-truth validation

OpenLandMap calibration. On the $n = 2,983$ WoSIS pedons, OpenLandMap SOC matches lab measurements at $R^2 = 0.547$ on $\log(1 + \cdot)$ (raw $R^2 = 0.633$), with a systematic bias of $+15.8$ g/kg (lab values higher than OpenLandMap) and $\text{RMSE} = 39.1$ g/kg (Figure 3). *The original paper’s $R^2 = 0.75$ was therefore alignment between two models, not a true accuracy measurement.*

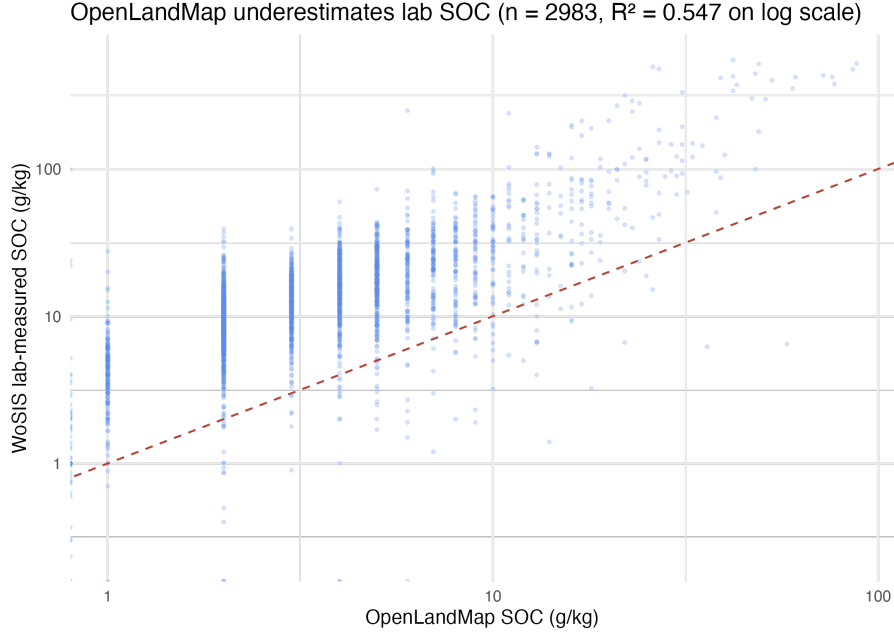


Figure 3: OpenLandMap SOC vs. WoSIS field-measured SOC at 2,983 points. Log-log scatter; dashed line is $y = x$. OpenLandMap systematically underestimates field SOC; $R^2 = 0.547$ on the $\log(1 + \cdot)$ scale.

Refit on field labels. Re-running R26’s recipe with WoSIS SOC as target yields $R^2 = 0.322$ (BCa 95% CI [0.273, 0.379]), RMSE = 0.679 on the log scale. The drop from 0.748 to 0.322 is the *label-noise floor*: WoSIS contains real measurement noise and sub-pixel heterogeneity that OpenLandMap smooths over.

Multi-target on WoSIS. The relative ranking of properties is preserved between OpenLandMap and field labels (Table 3), with pH again the most predictable property. This is robust evidence that the original paper’s per-target ordering reflects real signal rather than label-noise artefacts.

Table 3: Per-target spatial-block-CV R^2 on WoSIS labels, compared to the OpenLandMap result. Ranking preserved across both label sources.

Target	n_{WoSIS}	$R^2_{\text{OpenLandMap}}$	R^2_{WoSIS}
pH	2,889	0.896	0.539 [0.489, 0.582]
$\log(1 + \text{SOC})$	2,983	0.748	0.322 [0.273, 0.379]
$\text{logit}(\text{clay } \%)$	2,864	0.559	0.292 [0.207, 0.326]
$\text{logit}(\text{sand } \%)$	2,803	0.643	0.269 [0.231, 0.295]
Bulk density	644	0.806	0.161 [0.120, 0.189]

Spatial residual analysis. Moran’s I [10] on the $n = 2,000$ subsampled out-of-fold residuals gives $I = 0.104$ ($p < 10^{-15}$): residuals retain significant positive spatial autocorrelation. The 5-fold 1° -block CV attenuates spatial leakage but does not eliminate it. We report the R^2 as a defensible upper-bound for spatial generalisation; finer block sizes (e.g., 5°) would yield a more conservative number.

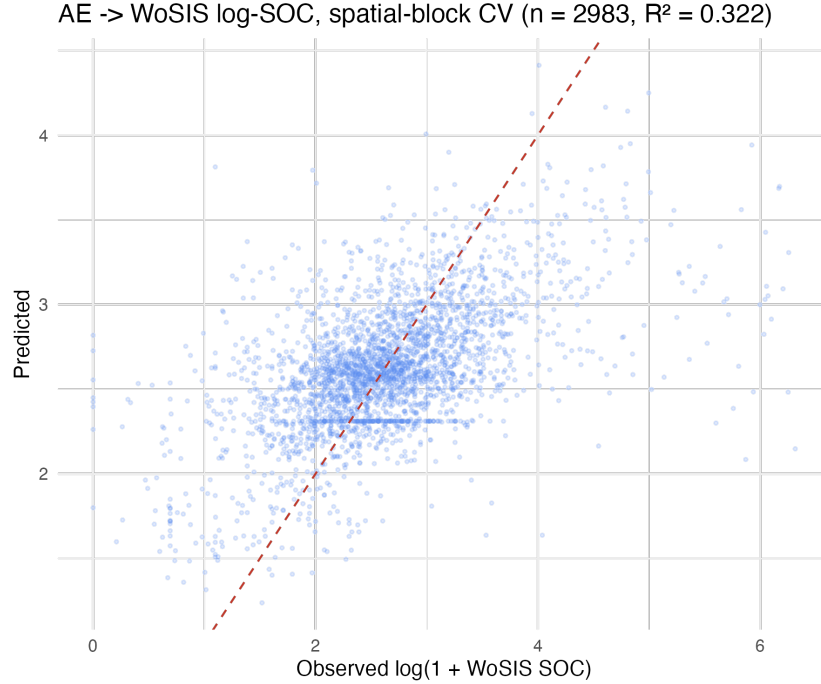


Figure 4: AlphaEarth $\rightarrow$ WoSIS log-SOC, out-of-fold predicted vs. observed across 5 spatial-block folds.

4 Discussion

Production implications for soilmetrix-platform. The benchmark is decisive: a foundation-model embedding adds 21.5 percentage points of R^2 over a representative open Sentinel-2 stack for SOC, while perfectly absorbing that stack’s signal. For the soilmetrix-platform’s existing `services/data-ml` ensemble (currently RF + GBM + XGB on NDVI-derived features), promoting AlphaEarth to a feature in `preprocessing.py` should produce a meaningful uplift. The multi-target result further suggests that the AE feature can serve all five soil properties the platform now supports, rather than requiring separate per-property feature engineering.

Honest scope. On field-measured labels the absolute R^2 is modest (0.32 for SOC). Two factors compound: WoSIS pedons span 1900–2020 against AE 2024 imagery (temporal mismatch), and a 10m pixel cannot resolve sub-pixel heterogeneity captured by a hand-augered profile. Both bound any remote-sensing approach, not just AE.

5 Limitations and Future Work

- Re-extract S2 with the bare-soil composite recipe of [12] once a less memory-bound execution path exists (per-tile export, then sample). Bare-soil compositing typically adds 0.05–0.15 R^2 over an annual median for SOC.
- Validate against USDA NCSS/KSSL pedons directly rather than WoSIS’s CONUS subset, especially post-2010 GPS-located dry-combustion samples.

- Multi-depth: refit at 0–30, 30–60, and 60–100 cm to test whether AE’s signal degrades with depth as expected pedologically.
- Replace the single Random Forest with the soilmetrix RF/GBM/XGB ensemble for a production-style benchmark.
- Repeat the ΔR^2 comparison at 5° spatial blocks to bound the effect of the residual spatial autocorrelation flagged by Moran’s I .

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